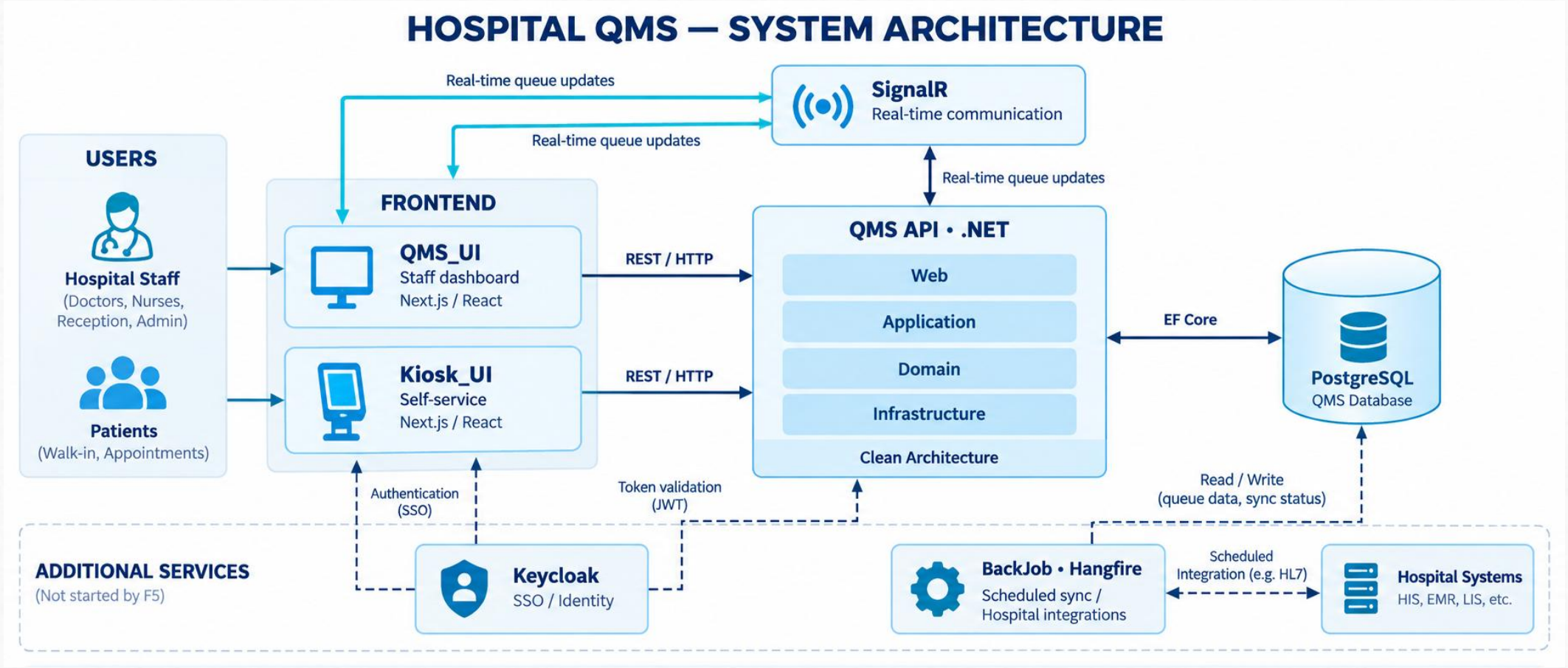
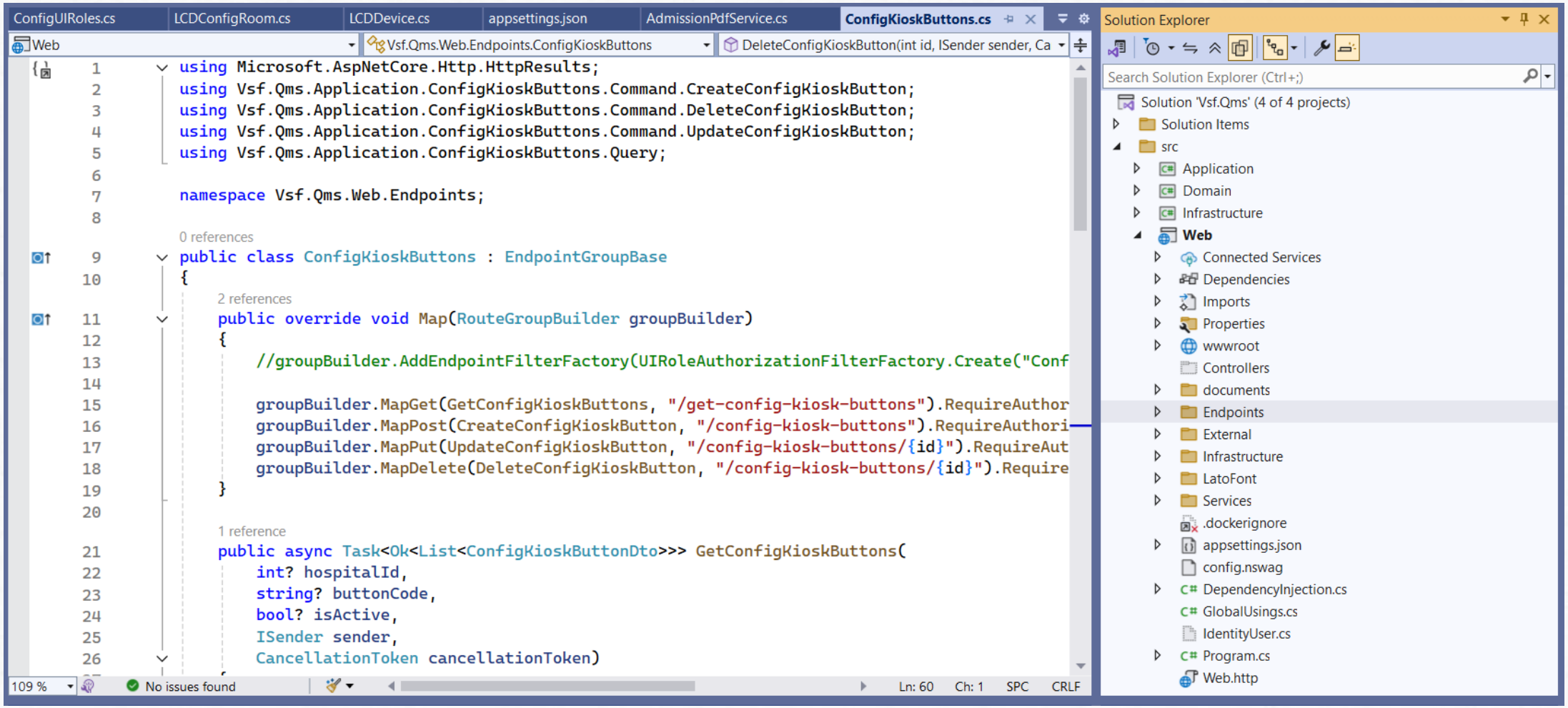
1. Clean Architecture: Systems

Hospital QMS manages patient queues and self-service registration through Next.js and React interfaces. A layered .NET API connects to PostgreSQL, while SignalR provides real-time updates. Keycloak supports centralized authentication, and Hangfire handles scheduled synchronization.



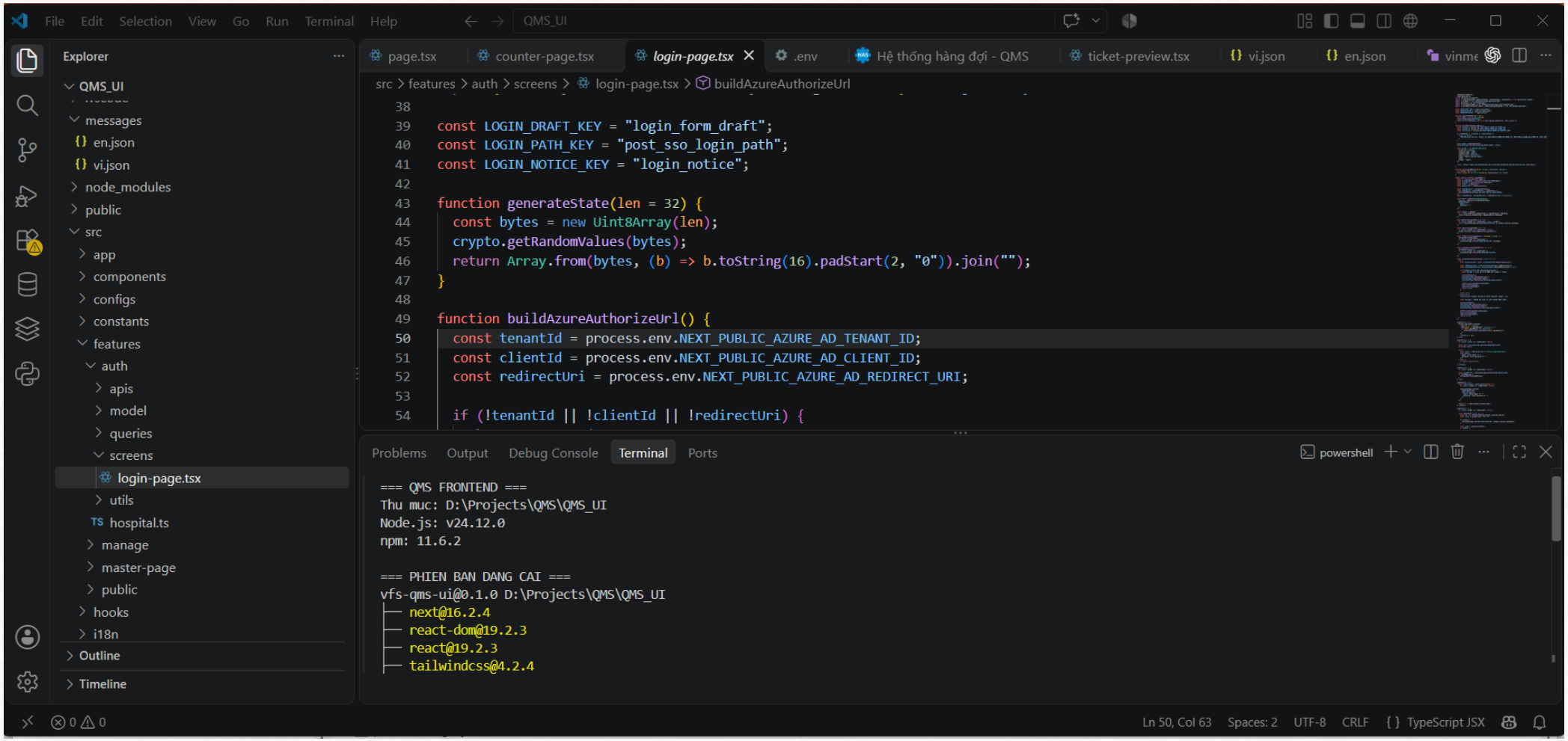
1. Clean Architecture: Backend Code Structure

The .NET backend uses Clean Architecture to expose secured endpoints for QMS configuration and operations.



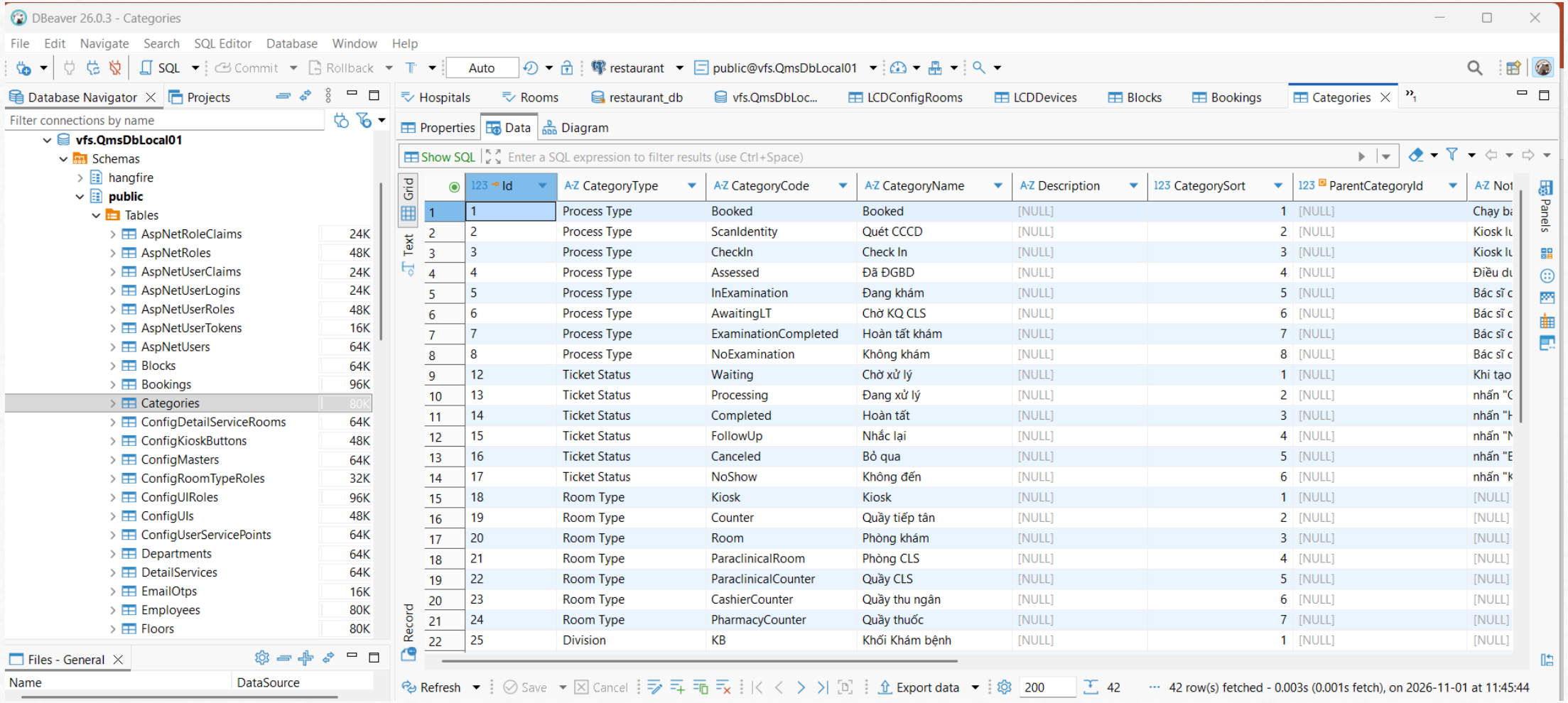
1. Clean Architecture: QMS Frontend Code Structure

The Next.js and React frontend organizes reusable modules for QMS, kiosk, and authentication workflows.



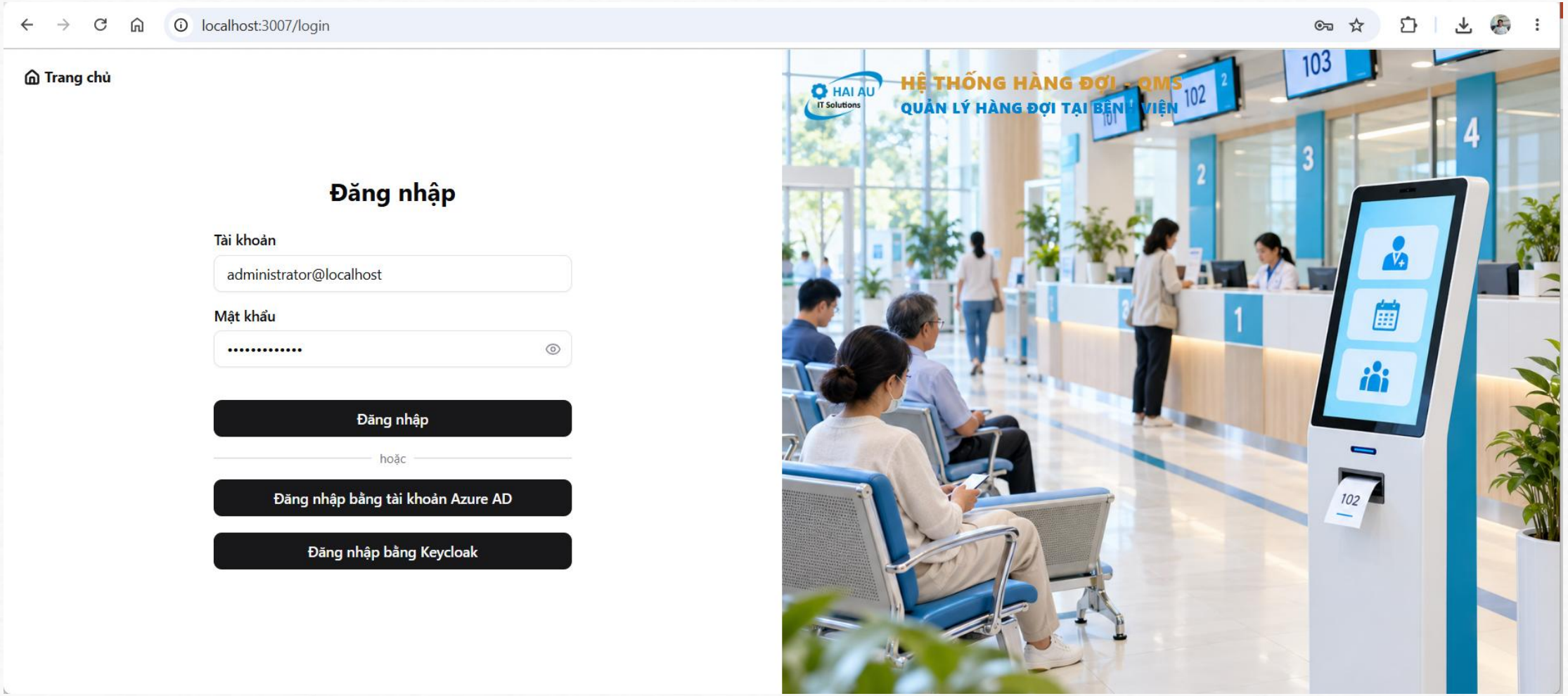
1. Clean Architecture: Database

PostgreSQL stores QMS configuration, rooms, bookings, users, queues, and permissions.



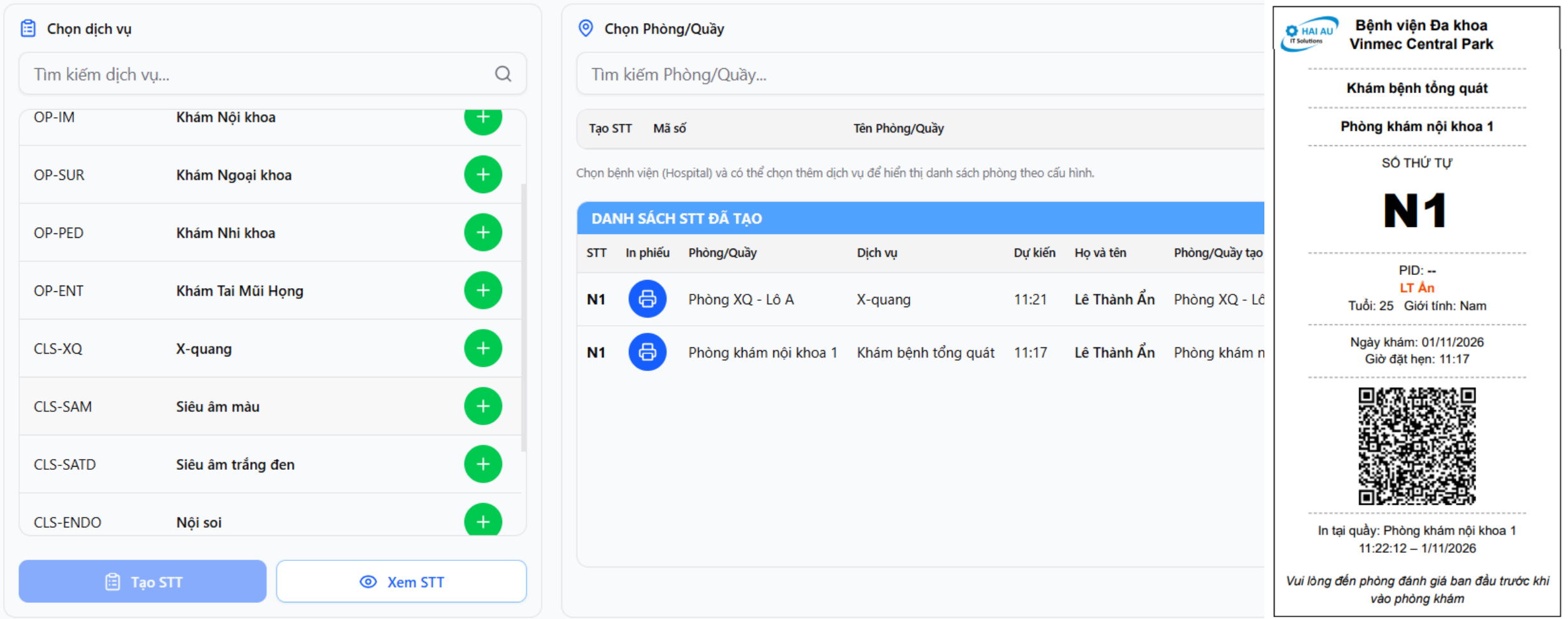
2. QMS: Login

The QMS login supports local accounts, Azure AD, and Keycloak authentication for hospital staff.



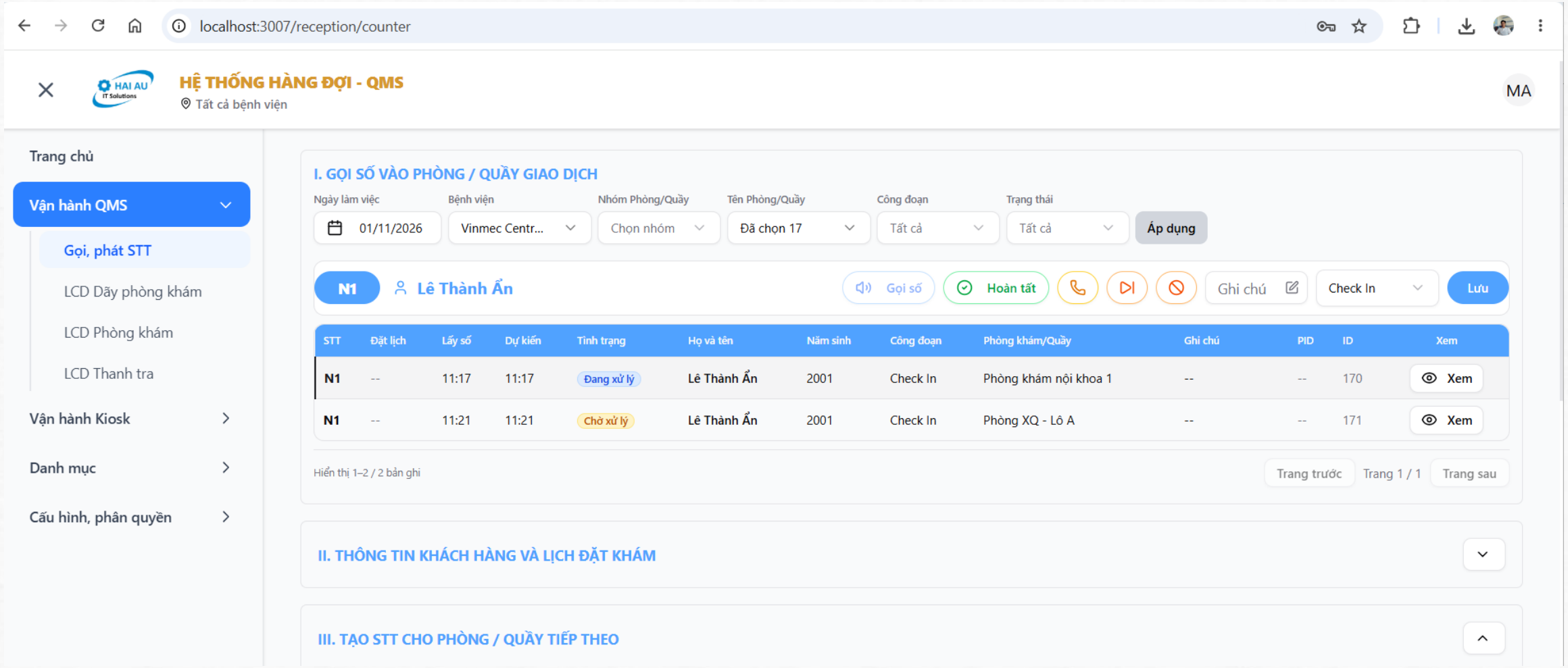
2. QMS: Check-in

The check-in screen selects services and rooms, creates queue numbers, and prints patient tickets.



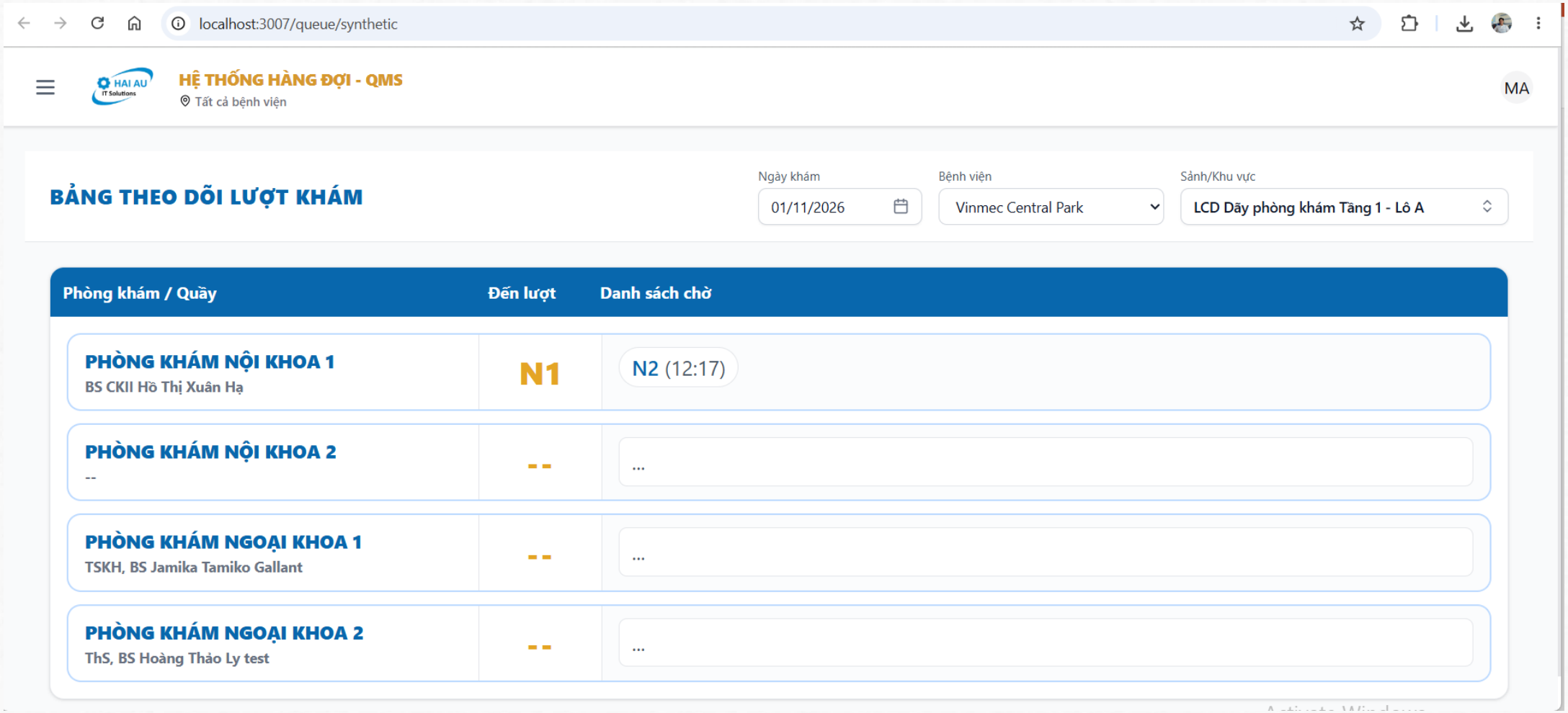
2. QMS: Call queue

Staff can call, recall, complete, skip, or transfer queue entries from one screen.



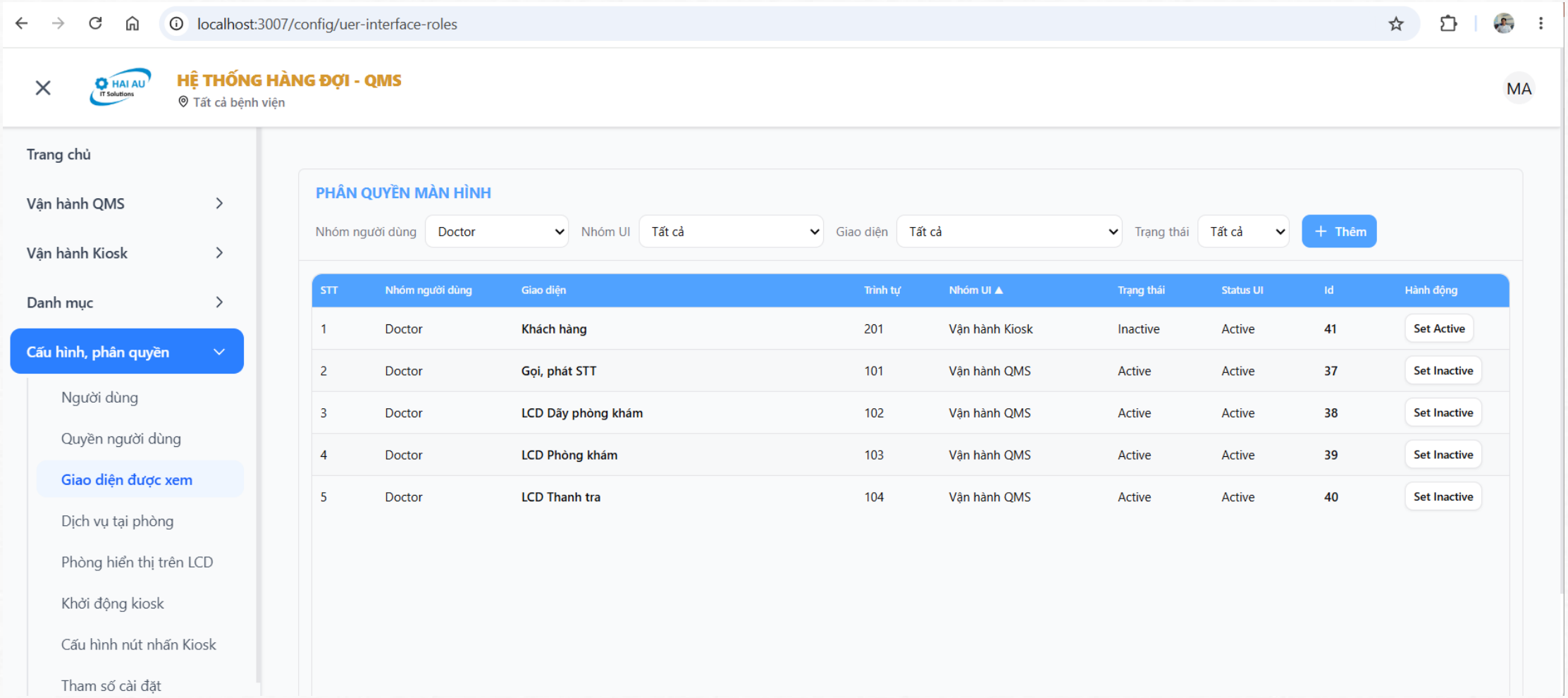
2. QMS: Display Queue at room

The room display shows the current queue number and waiting list for each examination counter.



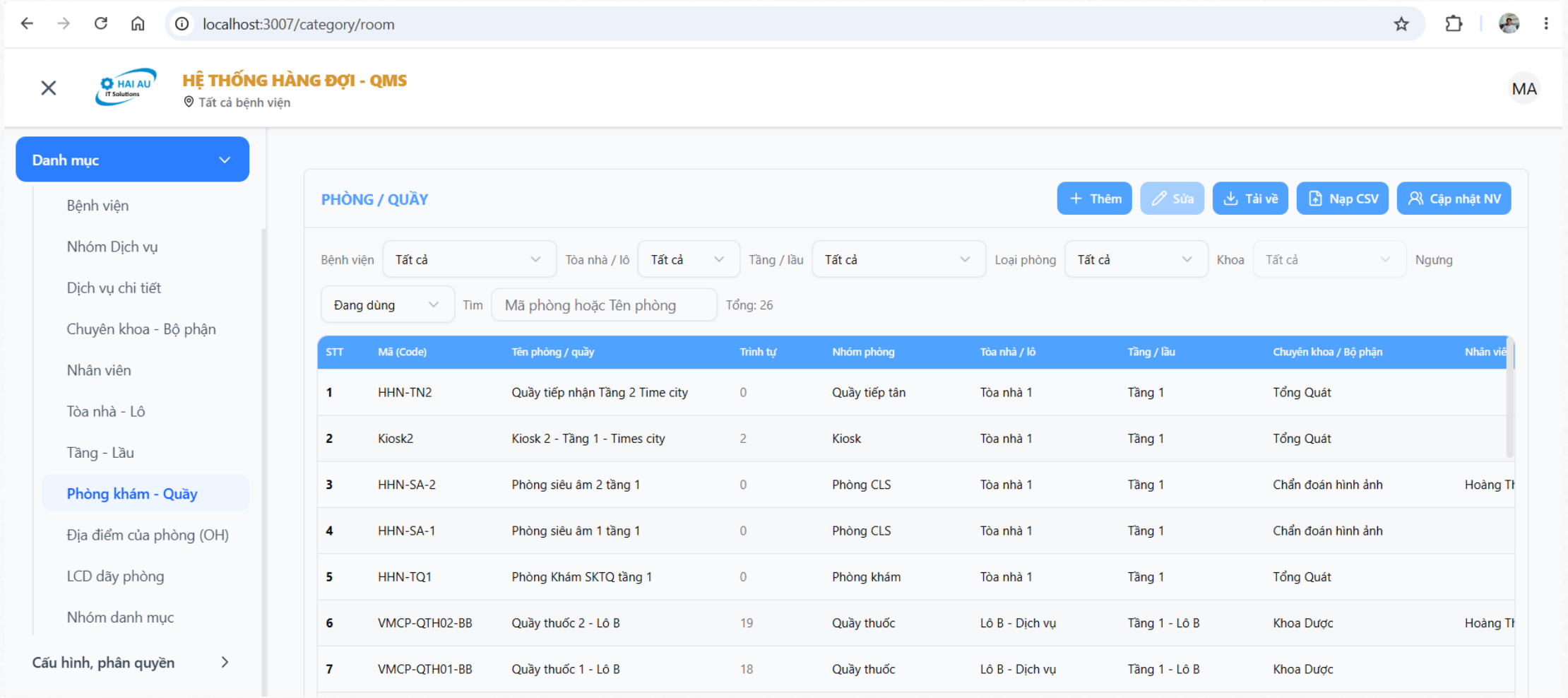
2. QMS: CMS Permission

CMS permissions control role access to QMS, kiosk, and room-display functions.



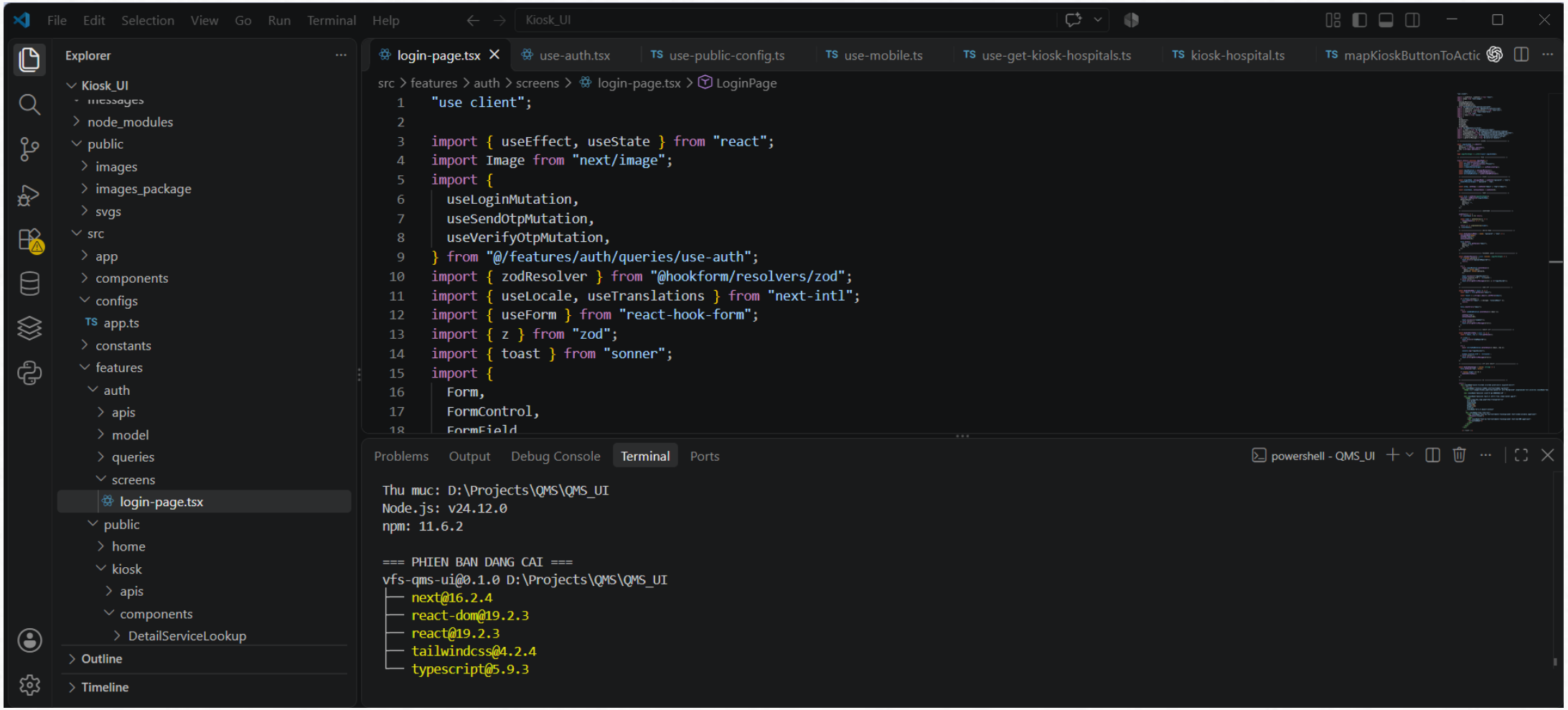
2. QMS: CMS Categories

CMS categories maintain rooms and counters with location, department, and status details.



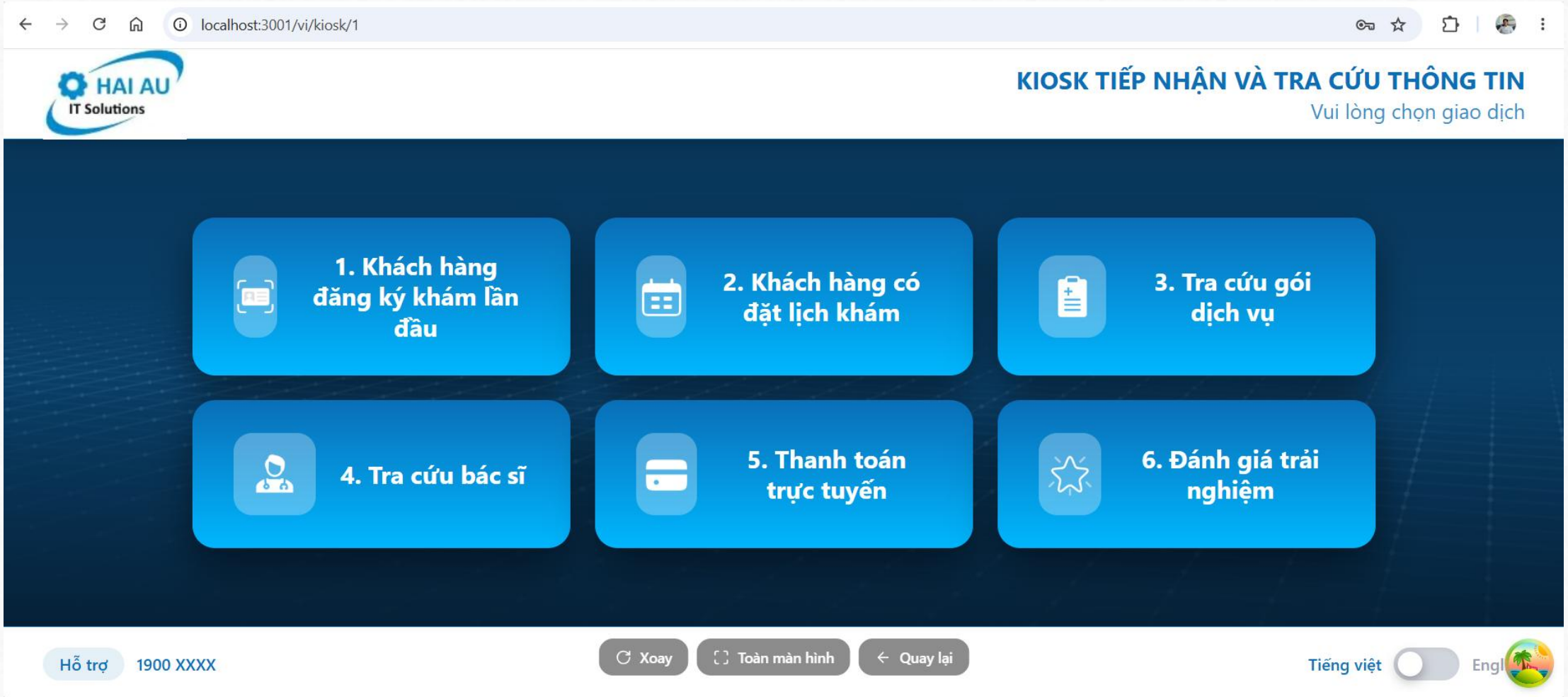
3. Kiosk: UI Frontend Code Structure

The kiosk frontend uses Next.js and React modules for multilingual patient self-service.



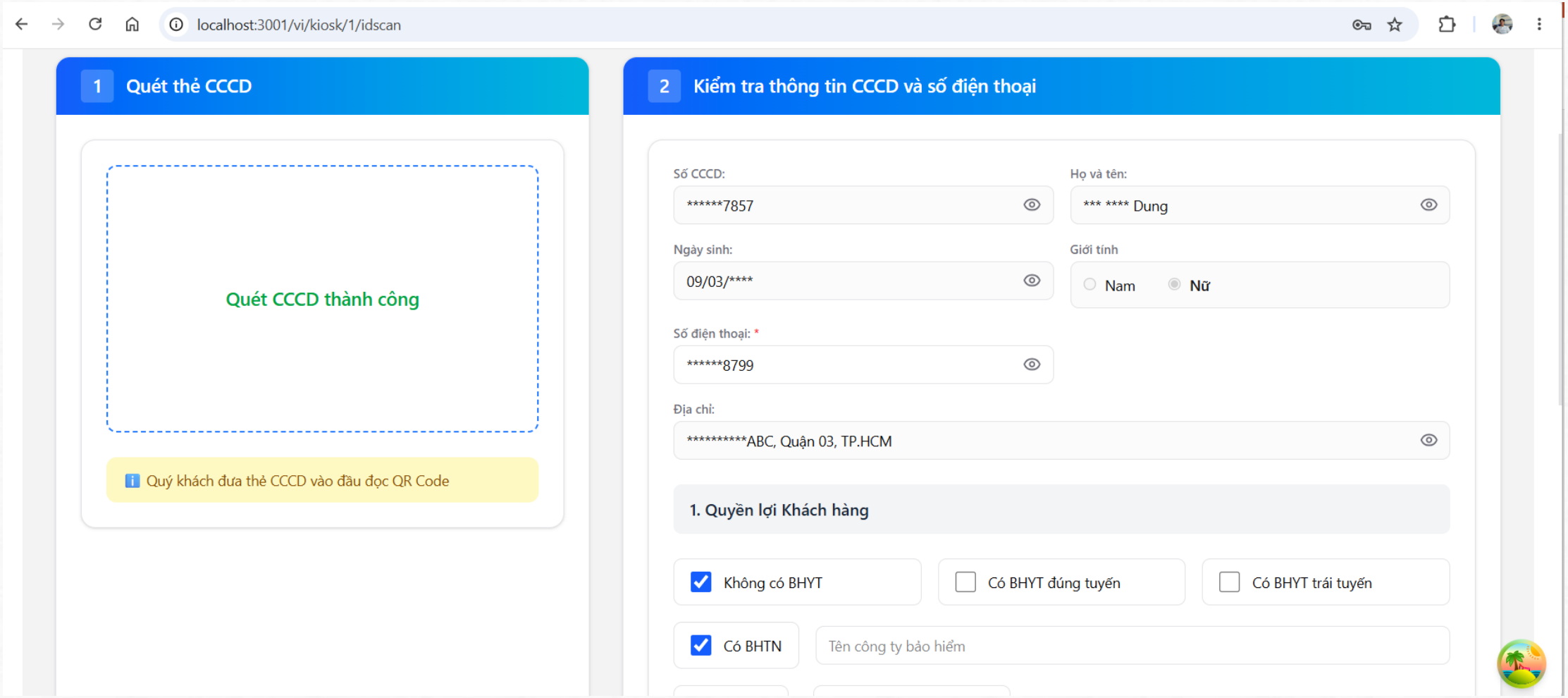
3. Kiosk: Home Monitor

The kiosk home screen provides registration, appointment lookup, service search, payment, and feedback.



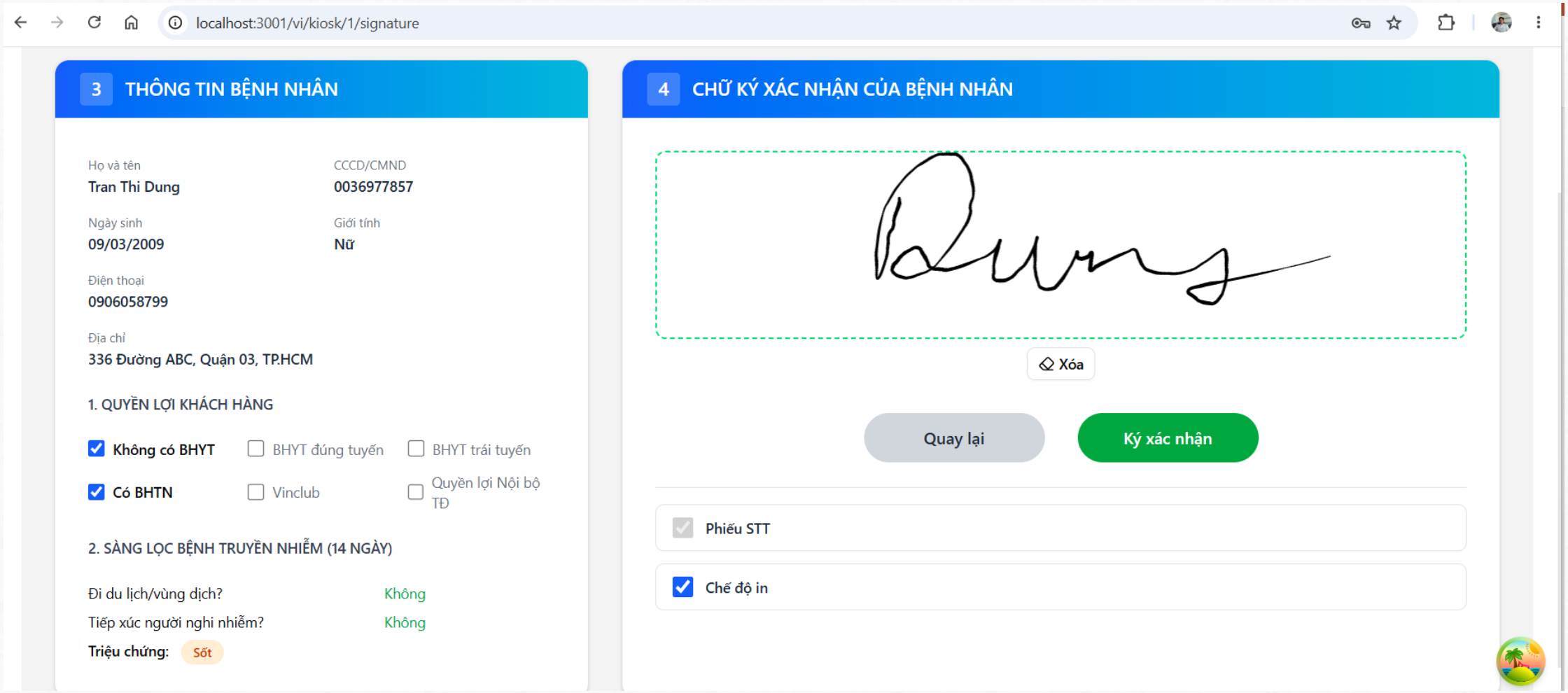
3. Kiosk: Scan Identity Card

The kiosk scans national ID cards and validates patient identity, contact, and insurance details.



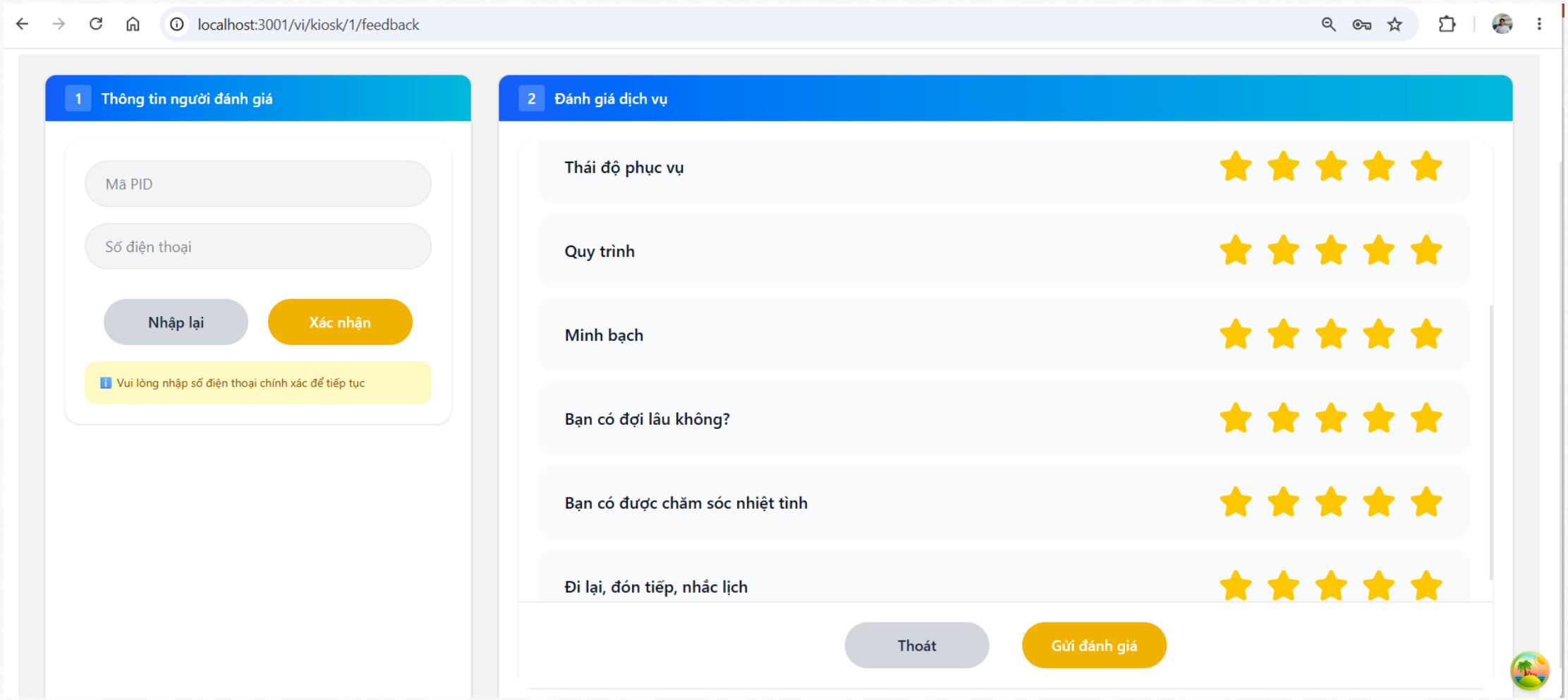
3. Kiosk: Sign and print queue ticket

Patients review their data, sign digitally, and request a printed queue ticket.



3. Kiosk: Research Questionaries

The kiosk captures patient ratings for service attitude, process, transparency, and waiting time.



3. Kiosk: Stored Data on CMS

CMS stores kiosk patient profiles and visit data for search, review, and administration.

